

Document Assessment: What Remains Relevant in docs/

Hermes Agent (Lt. Cmdr Data), for Prof. Dr. J. Kapeller

2026-09-14

Version 8 (September 2026) **Version 7** (September 2026) **Version 6** (September 2026) **Version 5** (September 2026)
Version 4 (September 2026) **Version 3** (September 2026) **Version 2** (September 2026) **Version 1** (September 2026)

This document is the working plan for the Metroeconomica revision: Foundation I (labour-market closures), Foundation II (financial closures), the 5 x 3 evaluation matrix, and the staged workplan with integrated reviewer-response maps. It evolved from the document assessment of Versions 1–2, whose chronological review now lives in docs/archive/document_review_leftovers.md. Governing documents: ROADMAP.md and roadmaps/vertdict.md; active references: definitive_guide.md, labor_closures.md, and docs/reviews/.

Version 5 reports the executed 5 x 3 matrix on the re-anchored A-bill calibration, the accounting correction that closed the external-identity canary, and the two closure fixes (real-wage supply, verified scale determinacy) that made the fixed-wage rows executable. Section 4.2 is new; where the earlier intermediate-results section disagrees with it, section 4.2 governs.

Version 6 separates real GDP from welfare in the measurement layer (ADR-0018). Real GDP is now the income-side index $w \cdot \sum(L)$ deflated by a seven-component Tornqvist GDP deflator, with the expenditure-side Divisia index as its dual and the external wedge reported as a diagnostic; the legacy household-consumption index is relabelled as welfare and is no longer reported as GDP. Because the kernel measurement layer changed, the matrix is re-minted as matrix_5x3_v4 (the same 15 cells and parameters as v3; v1-v3 stay as history, ADR-0004). Under the demand-only design the deflator is exactly one, so real GDP is flat (0.000 percent) in every full-employment mobile row: the F2 column shows the tax-financed programme as GDP-neutral with a 1.69 percent welfare loss, not as a GDP contraction. The v5 reading of the F2 column is corrected in section 4.2; the manifest-generated v4 flow tables are in paper/tables/matrix_5x3_v4_flows.md.

Version 7 harmonizes the external account (ADR-0019). All N goods-market clearings are now enforced in every regime: the mobile $\eta = 1$ system carries the endogenous net external transfer F with $E = (1 - \tau) w \sum(L) + F$, the $\eta = 0$ endpoint pins $F = 0$, and programme financing under F3 is booked as $B_{\text{gov}} = \sum(p g)$, so the booked identity $S + T_{\text{int}} + M - (I + X) = F + B_{\text{gov}}$ holds to about $1e-16$ at the mobile cells. Because the kernel changed, the matrix is re-minted as matrix_5x3_v5 (15 of 15 cells executed; v1-v4 stay as history, ADR-0004); the manifest-generated flow tables are in paper/tables/matrix_5x3_v5_flows.md. Financing neutrality is the generation's finding: F2 and F3 have identical real allocations in the mobile regime with $F_{\text{F3}} = F_{\text{F2}} - B_{\text{gov}}$ and an identical net external position, so the F2/F3 differences reported in v1-v4 were artifacts of the omitted-market shortcut, not economics. Measured movement against v4: mobile F1 welfare +0.043 percent becomes +0.143 percent, mobile F2/F3 welfare -1.694 percent becomes -1.823 percent, and the fixed rows and the BF rows are unchanged. Correction of the same date, within this version: the BF $\eta = 0$ external-position entries are booked financing rather than positions – the $F = 0$ pin reduces them to the programme cost $B_{\text{gov}} = +1.331$ percent of GDP, BF-F3's account is open by -0.7936 percent of GDP, and the $\eta = 0$ position is not identified (every pin $F = c$ is an exact root); see section 4.2 and ADR-0020 (accepted, option C, and promoted the same day: the matrix is re-minted as matrix_5x3_v6, whose BF row now has an identified external position and a closed account while the twelve non-BF cells reproduce v5).

Version 8 records the executed supply arm (ADR-0023): the plain channels the equivalence analysis left as an open route are now measured. Three designs — supply_etas_s1 (sector-1 shock, $A_1 = 1.1/1.2/1.3$), supply_etas_prog (the programme's own sectors, $A_i = 1 + \alpha \psi_i$, $\alpha = 0.05/0.10/0.20$), supply_etas_unif (uniform, $A_i = 1.01/1.02/1.03$) — 63 cells each, every cell executed with all gates passing. The identification result is exact and magnitude-invariant (implied $\ln L / \ln w = \text{input } \eta_s$ to $1e-3$, spread across the magnitude ladders below $1e-3$); the ALPHA control pins $L = 1$ ($w = 1.0049 / 1.0046 / 1.0408$); financing neutrality $F2 = F3$ holds under every supply shock; and the ADR-0019 external-account canary is asserted at $A \neq 1$ on all 189 cells — the ETAs gate that had only been tested at $p = 1$. Finding 2's supply-side parenthetical is thereby promoted from probe to executed generation (the upgrade note sits under finding 2). The programme size is settled at the deflated horizon mean (ADR-0024: $G0 = 40.3$ bn at 2019 prices = 1.331 % of GDP, i.e. the 2023-price mean 58.9 bn / 1.46 deflator). The revision palette restarts with the brighter cycle for Version 8 (revisionV7..V12, the same hues at higher brightness).

Version 4 reports the intermediate implementation results of the notebook pipeline `cbase2/` (v3 open-economy recalibration): the Stage 0 chapter is promoted to its own section with an updated status matrix, the remaining stages form the new Workplan section, and Stage 1 carries its updated state.

The document now opens with the two foundation sections whose decisions gate everything else – the labour-market closure (Foundation I) and the financing closure (Foundation II) – followed by the chronological workplan. All of this is part of the Version 2 reworking; the option tables in the foundation sections carry, per row, formulation, status, referee point, cost, and recommended role, plus the provenance tags (reviewer, old docs, Baqaee–Farhi usage).

1 Foundation I: Labour-Market Closure [added v2]

The referee demand (R2) is an explicit labour-supply elasticity; the code currently implements a geometric reallocation parameter and two wage regimes. The option tables below consolidate the full design space, grouped into the selected closures (BF plus the four standard alternatives ALPHA–DELTA), notable unselected candidates, and the dropped exogenous-shift closure. Provenance tags: "reviewer" attributes R1/R2; "old docs" names documents (archived under `docs/archive/`, plus the governing `ROADMAP.md` and `roadmaps/vertdict.md`); "used by BF" points to the Baqaee–Farhi papers (both source-verified in `bf_replication/` and `bf_replication2/`):

- BF 2019 (Econometrica): `Dokumente/Baqaee-2019-The Macroeconomic Impact of Microeconomic Shocks-Beyond Hulten's Theorem-Econometrica.pdf`
- BF 2022 (AER): `Dokumente/Baqaee-2022-Supply and Demand in Disaggregated Keynesian Economies with an Application to the COVID-19 Crisis-American Economic Review_4.pdf`

1.1 Selected closures [added v2]

Suggested narrative: the paper measures how far the estimated effects of a sector-specific green-investment programme travel between the CGE and IO endpoints as the labour-market closure varies. Closure BF operationalizes the neoclassical short-run argument as a market friction: BF 2019's own framework contrasts full reallocation with none, and the η parameter makes that friction continuous – limits on intersectoral reallocation, not a change of paradigm, separate the short run from the benchmark. On the shared production-network core, four standard closures span the space: ALPHA fixes aggregate labour at full employment (the neoclassical standard); BETA relaxes the endowment through a labour–leisure elasticity (the voluntary response the referees ask for); GAMMA fixes the real wage and lets employment absorb the shock (production-agnostic Keynesian: the mechanism operates on the labour market rather than through the production function, so the extensive-margin result survives any production structure); DELTA fixes the real factor price with unconstrained quantity and fixed coefficients, emulating the Leontief inverse (Robinson 2006). The measurement claim: the wage regime (ALPHA/BETA vs. GAMMA) and the substitution elasticities govern the distance to DELTA; the reallocation friction BF is material but secondary; and the financing closure keeps every experiment admissible.

Closure BF: Geometric reallocation parameter (η, market friction)	<i>Endogenous sectoral allocation between immobile and cost-minimizing; the BF-comparable short-run friction</i>
Formulation	$L_i = L_{i,\text{fixed}}^{1-\eta} L_{i,\text{costmin}}^\eta$ with $\eta \in [0, 1]$ (extrapolation outside); total employment fixed at $\bar{L}$; not a labour-supply elasticity
Status	Implemented and tested (<code>MobileLaborCES</code> , <code>:mobile</code>); Sobol first-order share about 15.7%
Referee point addressed	The mobility margin only; total supply stays fixed
Cost	None
Recommended role	Selected: the BF-comparable market-friction closure; the interpolation parameter is reported under an honest name ("reallocation parameter")
Mentioned by reviewer	both (R1.6 mobility/wage-rule question; R2.5 immobile-labour criticism – indirectly)
Mentioned in old docs	<code>labor_closures.md</code> ; <code>definitive_guide.md</code> ; <code>WORKPLAN3.md</code> (2026-09-03 addendum)

Closure BF: Geometric reallocation parameter (η, market friction)	<i>Endogenous sectoral allocation between immobile and cost-minimizing; the BF-comparable short-run friction</i>
Used by BF	Yes at its endpoints – the original BF 2019 replication ships both variants: immobile labour (“no reallocation”; sector-specific wage vector, <code>Simulation_Derivs.m</code>) and mobile labour (“full reallocation”; single wage pinned at the numeraire, <code>Simulation_Derivs_realloc.m</code> ; <code>Read_Me.txt</code> : “the first replicates results with immobile labor and the second replicates results with mobile labor”). The geometric interpolation parameter <i>between</i> the endpoints is project-specific
Closure ALPHA: Full-employment mobile labour (neoclassical standard)	<i>The standard full-employment CGE endpoint</i>
Formulation	$\sum_i L_i = \bar{L}$; one economy-wide wage; cost-minimizing allocation; the $\eta = 1$ limit of the reallocation parameter (as <code>:mobile</code>)
Status	Implemented (<code>:mobile</code> ; ROADMAP Closure A); baseline endpoint of the wage-regime sweep
Referee point addressed	R2.5 (economy-wide mobile labour); R1.6 (unified wage rule enabling mobility)
Cost	None
Recommended role	Baseline endpoint of the closure continuum
Mentioned by reviewer	both (R2.5 explicitly recommends it; R1.6 asks for it)
Mentioned in old docs	<code>REVISED_SALVAGE_PLAN.md</code> ; <code>WORKPLAN.md</code> ; <code>WORKPLAN2.md</code> ; <code>verdict.md</code> ; <code>definitive_guide.md</code>
Used by BF	Yes – BF 2019 baseline: competitive core with inelastically supplied (fixed) composite-factor endowment and uniform factor prices (see filename above)
Closure BETA: Elastic total labour supply (neoclassical, labour–leisure)	<i>Voluntary labour-supply response along the real wage; the continuum from vertical to horizontal supply</i>
Formulation	$L^s = \bar{L} [(w/P)/(w_0/P_0)]^{\eta_s}$ with a labour–leisure interpretation; $\eta_s = 0$ fixed supply; $\eta_s \rightarrow \infty$ approaches the fixed-real-wage limit
Status	Not implemented; specified in ROADMAP sections 4.2 and 5 (Closure B)
Referee point addressed	R2’s core demand (“the one obvious set of elasticities that really matters”)
Cost	Moderate: extends the $2N+1$ system by the supply equation and its anchor
Recommended role	Selected as the referee-facing principal addition
Mentioned by reviewer	both (R2 explicitly and centrally; R1 implicitly via the slack-scenario discussion)
Mentioned in old docs	<code>REVISED_SALVAGE_PLAN.md</code> (original <code>elastic_labor_slack</code> proposal); <code>WORKPLAN.md</code> ; <code>WORKPLAN2.md</code> ; <code>PRELIMINARY-ASSESSMENT.md</code> (baseline-anchoring correction); <code>verdict.md</code> (<code>ElasticLaborSupply</code>); <code>ROADMAP.md</code>
Used by BF	No (2019: inelastic endowments; 2022: sticky wages – <code>verdict</code> names the 2019 base as the right foundation for adding it)

Closure GAMMA: Sticky real wage, uncapped (production-agnostic Keynesian)	<i>Fixed real wage; employment absorbs the shock – the extensive margin</i>
Formulation	$w/P = \bar{\omega}$ (wage pinned at baseline); employment endogenous and uncapped; the gap $\bar{L} - \sum_i L_i$ is computed post-solve, not an equilibrium constraint
Status	Implemented; large positive GDP responses (+19.3 pp at mult 10, employment to 1.19x), residuals at most 1.9×10^{-7}
Referee point addressed	R1.5 (wage-setting rule: fixed real/nominal wage); R2’s “fix the wage rate at baseline and drop the labour supply constraint” recipe
Cost	None
Recommended role	Report as-is, with the uncapped-employment caveat stated
Mentioned by reviewer	both (R2 gives the recipe explicitly; R1.5 demands the wage-setting rule)
Mentioned in old docs	WORKPLAN3.md (Milestone F + 2026-09-03 addendum); definitive_guide.md (Result 3); labor_closures.md
Used by BF	Yes (adapted) – BF 2022’s sticky-wage mechanism (downward wage rigidity, demand-determined employment) in simplified one-wage form (see filename above)
Closure DELTA: IO endpoint (emulating the Leontief inverse)	<i>Fixed real factor price, unconstrained factor quantity; the exact conditions under which the CGE reproduces the IO multiplier</i>
Formulation	Fixed real primary-factor price; unconstrained factor quantity within the experiment; Leontief input coefficients (or a demonstrated CES special case with unchanged relative prices)
Status	Not implemented as an explicit closure (ROADMAP Closure D); GAMMA approximates its spirit
Referee point addressed	R2 explicitly (Robinson 2006: fix primary input prices, unconstrained supplies – the multiplier model); R1 explicitly (McGregor–Swales–Yin long-run zero-price-change IO-type result)
Cost	Small–moderate
Recommended role	Required: R2’s entire critique turns on the exact conditions of equivalence
Mentioned by reviewer	both (R2 with the Robinson 2006 quote; R1 with McGregor et al. 1996)
Mentioned in old docs	verdict.md (FixedRealWage endpoint); ROADMAP.md (Closure D); definitive_guide.md (Part IV); REVISED_SALVAGE_PLAN.md (the superseded $\eta \rightarrow \infty$ endpoint idea)
Used by BF	No (origin is the IO tradition and Robinson 2006, not the BF papers; BF 2019 computes efficient reallocations, not fixed-price multipliers)

1.2 Notable candidates, but unselected [added v2]

Structural alternatives that did not make the selection: the partial-mobility 3N system (allocation-axis alternative) and the unemployment complementarity (optional, policy-evaluation only).

Standard partial-mobility formulation (3N system)	<i>Structural alternative: sectoral wages with wage-responsive sectoral supply</i>
Formulation	$L_i^s = \bar{L} s_i^0 (w_i/\bar{w}_i)^\nu / \sum_j s_j^0 (w_j/\bar{w}_j)^\nu$; sectoral labour demand from Shephard's lemma; N zero-profit + $(N-1)$ market-clearing + N sectoral labour-market + one numeraire = $3N$ equations
Status	Open design proposal in <code>labor_closures.md</code> ; not implemented
Referee point addressed	None directly; referee-adjacent (R1.6, R2.5)
Cost	Largest: a new equilibrium system
Recommended role	Fallback if the one-wage supply formulation of Closure B is judged too reduced-form
Mentioned by reviewer	no
Mentioned in old docs	<code>labor_closures.md</code> (the proposal, with references: Hsieh–Klenow 2009; Baqaee–Farhi 2020 QJE; Artuc–Chaudhuri–McLaren 2010)
Used by BF	No (closest analogue: BF 2022's N sector-specific labour markets, but those are sticky, not supply-responsive)
Closure C: Unemployment complementarity (capped)	<i>Employment capped at $\bar{L}$; full-employment or wage-floor regime switches endogenously</i>
Formulation	$0 \leq \bar{L} - L \perp w/P - \bar{\omega} \geq 0$: complementarity between the employment gap and the real-wage gap
Status	Not implemented (ROADMAP section 5, Closure C; definitive guide Part IV records “partially”)
Referee point addressed	R1.5 (wage-setting rule, wage curve, involuntary unemployment); R2 (DDR/Robinson unemployment closures; skill-class split)
Cost	Larger: complementarity solver, new results
Recommended role	Optional; mandatory only if the paper claims policy evaluation (ROADMAP Phase 6)
Mentioned by reviewer	both (R1.5 wage curve; R2's closure-literature demands)
Mentioned in old docs	<code>verdict.md</code> (UnemploymentComplementarity); <code>ROADMAP.md</code> ; <code>definitive_guide.md</code> (Part IV status)
Used by BF	Yes (adapted) – BF 2022's complementarity machinery for binding wage floors / labour capacity, to be stripped down; its N -sector sticky-labour structure must not be copied unchanged (verdict; see filename above)

1.3 Dropped [added v2]

Regretfully retired: the exogenous labour-endowment shift in its legacy-callback implementation – the original paper's own mechanism. The lineage note below records its direct resonance with BF 2022 and with the response letter's methodological story.

Dropped: Exogenous labour-endowment shift (the legacy labour-slack callback)	<i>The original paper's own mechanism: exogenous sectoral slack vectors – the same closure BF 2022 calibrates as labour-supply shifts</i>
Formulation	Sectoral labour endowments shifted exogenously: $\bar{L}_i \rightarrow \bar{L}_i(1 + g_i)$ with calibrated or assumed g_i ; no wage response required. Code implementations: <code>full_labor_slack</code> / <code>full_labor_slack_alt</code> (Leontief-projection sectoral vectors, manuscript Section 5.1) and <code>empirical_labor_slack</code> (uniform $\mathbf{l}_{\text{new}} = (1 - \mu)^{-1} \mathbf{l}$, Section 5.2)

Dropped: Exogenous labour-endowment shift (the legacy labour-slack callback)	<i>The original paper's own mechanism: exogenous sectoral slack vectors – the same closure BF 2022 calibrates as labour-supply shifts</i>
Status	Implemented (legacy); the original submission's Sections 5.1–5.2; ROADMAP section 4.4 lists it as a distinct closure concept
Referee point addressed	R1.5 describes precisely this mechanism (“is it simply a matter of increasing labour supply by the number of unemployed persons?”); R2 criticizes its use as the central bridge (deus ex machina; R2.10 uniform allocation)
Cost	None
Recommended role	Dropped from the model; its economics survive in the lineage note below and in the response letter's methodological story
Mentioned by reviewer	both (R1 describes the mechanism; R2 criticizes its use as deus ex machina)
Mentioned in old docs	REVISED_SALVAGE_PLAN.md; WORKPLAN.md; WORKPLAN2.md; labor_closures.md; definitive_guide.md; WORKPLAN3.md
Used by BF	Yes – BF 2022 calibrates exogenous sectoral labour-supply shifts to BLS May-2020 sectoral hours (contractions; <code>bf_replication2/src/model.jl</code> applies them on the labour block); run in reverse, the same closure is the slack injection. BF 2019 shocks factor endowments in its general framework

Lineage note (original manuscript). The resonance is direct, not analogical. The original submission's Section 5 already framed labour slack as exactly this closure: “the introduction of such a labor slack by itself is to be conceived and interpreted as the advent of an exogenous shock ... a spontaneous expansion of the labor force.” Its legitimizing narrative was borrowed from BF 2022's channel: “the context of Corona-related lockdowns that partially required people to abstain from work due to social distancing regulations” (with BF 2022's minimum-wage unemployment explicitly set aside because it “does not relax capacity constraints”). The code implements the manuscript's two variants: `full_labor_slack` and `full_labor_slack_alt` (Leontief-projection-calibrated sectoral vectors; Section 5.1, aggregate employment +1.7%) and `empirical_labor_slack` (the uniform rule $\mathbf{l}_{\text{new}} = (1 - \mu)^{-1} \mathbf{l}$ with μ the ILO unemployment rate of 3.2%; Section 5.2, +3.1%). The referees' treatments map onto these two variants: R1 questions 5.1's rationale (“unsure about the rationale behind this scenario”) and 5.2's missing wage-setting rule; R2's “deus ex machina” targets the injection as the bridge mechanism and the uniform transfer as arbitrary (R2.10). BF 2022 runs the same machinery as calibrated contractions; the original manuscript ran it as calibrated expansions. The revision therefore changes not the closure's existence but its role: from the central bridge mechanism to a named scenario closure, with the bridge carried by endogenous labour supply (Closure B), the wage regime, and an explicit financing closure.

1.4 Coverage check: BF closures outside our lists [added v2]

Verified against the two papers ([Dokumente/](#)) and the replications; it spans both foundation lists. BF 2019 itself uses nothing beyond Closure A: its factor supply is inelastic at the endowment (“the supply of the factor is inelastic”), factors are non-reproducible goods, and the paper explicitly makes “no allowance for amplification of shocks through endogenous labor supply and capital accumulation” – elastic labour supply (Closure B) is named there as a channel they do not use. BF 2022, however, uses four closure elements that our two foundation lists do not cover:

BF 2022 element	Coverage status in our lists
Downward nominal wage rigidity, sector-specific (N sticky labour factors)	Partially covered by Closure C (adapted, stripped down). Two differences: nominal floor vs. our real-wage floor (moot only because the static model has $\text{CPI} = 1$; it bites under deflation), and N sector-specific floors vs. one economy-wide floor. The replication implements sticky sectoral labour but not the nominal/deflation side

BF 2022 element	Coverage status in our lists
Credit constraints / hand-to-mouth vs. Ricardian households	Not covered. Demand-side income-distribution closure (who spends the shock); our model has one representative household; ROADMAP leaves it out of scope
Zero lower bound / monetary policy	Not covered. Nominal closure; absent from the local replication (which keeps only nominal GDP accounting) and outside our static real model; relevant only if the paper’s question shifts, per the verdict’s 2019-vs-2022 criterion
Multiple primary factors (capital alongside labour; flexible capital vs. sticky labour in BF 2022)	Not covered by a foundation table. Our model and the 2019 replication are one-factor (value added as composite); ROADMAP section 5 carries it as the “optional extension” (mandatory only for policy-evaluation claims)
Firm capacity constraints (social distancing)	Not covered. BF 2022 models a supply shock as a labour-supply shift or a firm capacity constraint (“firms could be forced to operate at lower capacity ... due to social distancing”) – a rationing element outside our price-based static system
Reallocation frictions (adjustment costs, kappa)	Not covered by a foundation table. BF 2019 Online Appendix 2 (“Stuck Intermediates and Adjustment Costs”, the kappa variants); ours is a static model – <code>labor_closures.md</code> names convex employment-adjustment costs for a dynamic extension

None of these gaps blocks the current plan: the first is exactly the “stripped-down 2022 closure” the verdict prescribes, and the remaining three enter only via the optional extensions already flagged in Stage 0 (scope decisions) and ROADMAP section 5. They are listed here so the response letter can state precisely which BF elements are and are not used. One replication-code nuance to verify before citing the BF 2019 mobile variant as Closure A exactly: in the MATLAB code that variant pins $w = 1$ (numeraire) and solves only the N price equations, so the aggregate labour constraint is implicit rather than imposed; the paper itself states inelastic aggregate factor supplies. The exogenous labour-endowment shift, though present in both BF papers, is a different closure from the sticky wage: it moves supply without a wage response, whereas `:fixed` moves the wage floor and lets employment respond.

2 Foundation II: Financial Closures [\[added v2\]](#)

The four admissible experiment types of ROADMAP Phase 2. The current `demand_shock` is an unfinanced add-on and inadmissible by the ROADMAP’s own rule until one principal and one robustness variant are chosen; the explicit demand anchor introduced with the financing closure is also expected to resolve the $\eta = 1$ stall (scale indeterminacy) observed from multiplier 0.5 upward. Neither BF paper supplies a fiscal financing closure (verdict); only the compositional-shifter normalization has a BF 2022 analogue.

2.1 Selected [\[added v2\]](#)

The three financing variants carried into the evaluation matrix: the preference-reallocation/tax-financed bracket as principal, external debt as the third variant.

Budget-neutral preference reallocation	<i>Compositional household demand shift with the budget closed by construction</i>
Budget identity	$\tilde{\beta}_i = \beta_i d_i / \sum_j \beta_j d_j$; $Z(p) = \sum_j \tilde{\beta}_j p_j^{1-\sigma}$; $c_i^h = E^h \tilde{\beta}_i p_i^{-\sigma} / Z(p)$, so $\sum_i p_i c_i^h = E^h$ holds exactly
Economic interpretation	Compositional demand shift; the deduction runs through the household’s consumption composition (other categories shrink within the fixed E^h); government spending untouched; must not be described as an autonomous investment multiplier (verdict warning)
Cost / status	Low: normalization is half-done already (Milestone A normalized household demand)

Budget-neutral preference reallocation	<i>Compositional household demand shift with the budget closed by construction</i>
Recommended role	Robustness / diagnostic benchmark
Mentioned by reviewer	no (neither referee demands it; R1's demand-shock rationale question is adjacent)
Mentioned in old docs	verdict.md (redesign section 1); ROADMAP.md (Phase 2); definitive_guide.md
Used by BF	Yes (analogous treatment) – BF 2022 renormalizes compositional demand shifters so the budget holds exactly (bf_replication2/src/model.jl:442); but as a demand shifter, not a financing device (see filename above)
Tax-financed public investment	<i>Autonomous green investment with an explicit tax counterparty</i>
Budget identity	$\sum_i p_i g_i = T$ with a lump-sum or specified tax rule; separate government/investment vector g_i
Economic interpretation	Autonomous green investment financed by taxation
Cost / status	Medium: new g_i vector plus budget equation; not started
Recommended role	Principal (recommendation)
Mentioned by reviewer	no (the financing gap was identified internally; the referees demand shock transparency only)
Mentioned in old docs	verdict.md; ROADMAP.md (Phase 2); PRELIMINARY-ASSESSMENT.md (financing admissibility); definitive_guide.md (Part V)
Used by BF	No (verdict: neither model supplies the fiscal financing closure)
Debt/foreign-financed net expenditure	<i>Demand expansion with an intertemporal or external counterpart</i>
Budget identity	$\sum_i p_i g_i = B + F$ with a defined saving–investment or current-account rule
Economic interpretation	Net new expenditure against borrowing or the external balance. Static-horizon nuance: debt finance (B) is resource-equivalent to the lump-sum tax (the household's current consumption gives up the same resources; the distinction becomes real only in a dynamic setting), while foreign finance (F) is genuinely distinct – resources arrive via the external balance (imports serve the investment)
Cost / status	Medium–high: additional institutional balance
Recommended role	Third variant (external debt); B collapses into the tax variant at this horizon
Mentioned by reviewer	no
Mentioned in old docs	verdict.md; ROADMAP.md (Phase 2); definitive_guide.md
Used by BF	No

Financing design (user decision): taking the housing stimulus as vantage point and ruling out unfinanced manna, the realistic financing lies between the preference-reallocation and tax-financed variants, so the principal experiments are a bracket of the two. A third variant with external debt (foreign financing, F) is added: within this static real model it is the closest representation of the endogenous-money intuition that a borrowed euro need not be withheld from domestic consumption ex ante – the purchasing power enters, and the ex-post identity $I = S$ (here: the external deficit) records where the resources came from. The model itself cannot represent endogenous money: with the nominal side collapsed it displays only the ex-post accounting, so the F variant is an accounting open-economy closure, not a monetary mechanism. Imports feedback: at the current pipeline the imported content of the g_i vector is fixed by sector import

shares, so the external variant absorbs the demand through net imports mechanically; a behavioral feedback (relative prices shifting import shares) would require an Armington-type extension, for which the existing $\Omega_{\text{dom}}/\Omega_{\text{raw}}$ split is the natural hook. Deduction routes recap: preference reallocation deducts from the household's consumption composition, expenditure switching from old government expenditure, tax financing from household income. Recommendation: run the preference-reallocation/tax-financed bracket as principal, foreign financing as the third variant; debt (B) collapses into the tax variant at this horizon; expenditure switching remains available as a secondary option. The full 5 x 3 evaluation matrix is assembled in the next section.

Financing design (user decision): taking the housing stimulus as vantage point and ruling out unfinanced manna, the realistic financing lies between the preference-reallocation and tax-financed variants, so the principal experiments are a bracket of the two; external debt is added as the third selected variant. Deduction routes recap: preference reallocation deducts from the household's consumption composition, expenditure switching from old government expenditure, tax financing from household income via the lump-sum tax. Imports feedback: at the current pipeline the imported content of the g_i vector is fixed by sector import shares, so the external variant absorbs the demand through net imports mechanically; a behavioral feedback (relative prices shifting import shares) would require an Armington-type extension, for which the existing $\Omega_{\text{dom}}/\Omega_{\text{raw}}$ split is the natural hook. The full 5 x 3 evaluation matrix is assembled in the next section.

2.2 Dropped [added v2]

Regretfully retired as a separate variant: expenditure switching. Pure debt financing (B) is not given its own variant at all – within the static horizon it is resource-equivalent to the tax variant (the distinction becomes real only in a dynamic setting), so it would only re-label the principal tax experiment.

Expenditure-switching investment	<i>Reallocation within a fixed public budget</i>
Budget identity	$\sum_i p_i g_i = T$ with T raised by cutting other public spending
Economic interpretation	Public budget reallocated across sectors; the deduction runs through old government expenditure ; household budgets untouched; no new purchasing power
Cost / status	Low-medium
Recommended role	Robustness alternative
Mentioned by reviewer	no
Mentioned in old docs	vertdict.md; ROADMAP.md (Phase 2); definitive_guide.md
Used by BF	No

The endogenous-money issue. The motivation for the external-debt variant is the endogenous-money intuition: in a world where credit money is created alongside the loan, a borrowed euro need not be withheld from domestic consumption ex ante; the ex-post identity $I = S$ holds by accounting, but it does not describe an ex-ante fund that had to be taken from anyone. The static real model cannot represent this mechanism itself: with the nominal side collapsed there is no credit creation, no banking sector, no interest-rate channel – the model displays only the ex-post accounting. What the F variant does capture is the shadow of the intuition: purchasing power enters without an ex-ante sacrifice of domestic consumption, and the external deficit records where the resources came from. It should therefore be sold as an accounting open-economy closure, not as a monetary mechanism. The contrast between the tax-financed and external-debt variants is nonetheless paradigmatically productive: it operationalizes, inside one framework, the difference between a loanable-funds reading (domestic saving must finance investment ex ante) and an endogenous-money reading (finance precedes saving; the balance closes ex post) – exactly the kind of commensurability the revised paper set out to measure.

3 The Evaluation Matrix (5 x 3) [added v2]

Decision (user): three financial closures – the preference-reallocation and tax-financed bracket plus the external-debt variant – crossed with the five selected labour closures gives a 5 x 3 matrix of 15 model variants to evaluate. This is not overblown: each cell is one solve of a 71-sector static system, computationally trivial; the interpretive burden is managed by reporting one headline set per cell (Tornqvist real GDP response, employment, external balance) with sectoral detail in an appendix. Two design amendments: BF's row uses the pre-registered endpoints $\eta \in \{0, 1\}$ (the interpolated η^* is retired, ADR-0010), and DELTA is a corner of the matrix rather than an independent equilibrium row – its cells are reached as GAMMA plus Leontief technology. The matrix serves the paradigmatic main point directly: it turns the search for commensurability into a measured map, with financing closures setting the scale of the stimulus and labour closures its transmission margin.

Figure: Paradigmatic directions towards the short run. The horizontal axis is the labour-supply-elasticity continuum (ALPHA at zero, BETA intermediate, DELTA at the IO limit); BF is the orthogonal reallocation-friction dimension anchored at the fixed-supply end; GAMMA approaches the IO endpoint from the wage-rigidity direction (GAMMA plus Leontief technology). Drafted after the user's sketch.

Figure 1: Figure: Paradigmatic directions towards the short run. The horizontal axis is the labour-supply-elasticity continuum (ALPHA at zero, BETA intermediate, DELTA at the IO limit); BF is the orthogonal reallocation-friction dimension anchored at the fixed-supply end; GAMMA approaches the IO endpoint from the wage-rigidity direction (GAMMA plus Leontief technology). Drafted after the user's sketch.

Expected signatures, pre-registered before any Stage 2 run (headline quantities per cell; sectoral detail deferred):

Labour closure	Financing: preference reallocation	Financing: tax-financed	Financing: external debt F
BF (friction, η^*)	Composition shift under friction: aggregate $\sim$ flat; sectoral mix most distorted	Tax withdrawal with frictional reallocation; GDP response smaller than ALPHA	Externally financed demand, frictional allocation; import leakage plus misallocation
ALPHA (full employment)	Composition only; aggregate ~ 0 (the η -invariance diagnostic); sectoral reallocation	Crowding-out via wages/prices; aggregate ~ 0 (classic full-employment CGE)	Demand leaks to imports; wage rise moderated; external deficit = F
BETA (elastic supply)	Composition only; employment $\sim$ flat along the supply elasticity	Positive but modest; employment and wage share the adjustment (η_s -dependent)	Positive; import leakage dampens the domestic employment response
GAMMA (sticky real wage)	Composition shift; small positive employment response	The extensive-margin result: employment absorbs the shock (current: +19.3 pp at mult 10)	Employment absorbs the domestically-produced share; imports take the rest
DELTA (IO endpoint, corner)	IO multiplier on the compositional shift (value-added effects)	IO multiplier net of the tax withdrawal	Full IO multiplier with import-adjusted inverse; external deficit = F

Measured values for every cell are in section 4.2. Three readings of the table above need correction: the BF row's label carries the retired interpolated η -star (the endpoints $\eta \in \{0, 1\}$ are what runs); the parenthetical "current: +19.3 pp at mult 10" is a retired UNFINANCED result; and external deficit = F describes the gross external inflow, not the net position the model reports (section 4.2; the decomposition is in [paper/tables/matrix_5x3_v3_flows.md](#)).

Limit relations across the matrix (reported as results, not hidden as redundancy): BF at $\eta = 1$ coincides with ALPHA, and BF at $\eta = 0$ is the immobile benchmark; BETA's high η_s limit approaches GAMMA's fixed-real-wage regime; and GAMMA plus Leontief technology is DELTA. DELTA is therefore a corner, not an independent row; BF's endpoints duplicate ALPHA and the immobile benchmark. The coincidences are the bridge result: the same endpoint is reached from different directions, which is precisely the paradigmatic-geography claim of the revised paper.

4 Stage 0: Conceptual decisions [\[reworked v3\]](#) [\[section 4 since v4\]](#)

Resolved in this round: the labour-closure selection (Foundation I: BF, ALPHA, BETA, GAMMA, DELTA), the financing design (Foundation II: the preference-reallocation/tax-financed bracket as principal, external debt as the third selected variant; expenditure switching dropped, pure debt subsumed by the tax variant), and the consolidation of the exogenous-shift/legacy closure as dropped. The status matrix below completes the Stage 0 answer by locating every cell of the evaluation matrix in the current implementation. Two items remain open: the audit-trail disclosure decision (Stage 4) and the scope locks (skill classes, open-economy depth, Stream C off the critical path).

The two paradigmatic wedges. The matrix has two axes, and both are paradigmatic wedges. The first – the labour-market closure – is the referees' axis: it sets the transmission margin of a demand stimulus, from allocation friction (BF) through full employment (ALPHA) and voluntary supply (BETA) to wage rigidity (GAMMA) and the IO endpoint (DELTA). The second – the financing closure – runs through the financing assumptions: it sets where the purchasing power of the stimulus

comes from and through which budget it closes (household composition vs. household income vs. external balance), i.e. the loanable-funds versus endogenous-money contrast. Recommended strength of the second dimension: first-class as structure, bounded as claim. It is a full matrix axis and appears in every results table; but in the static real model it operates as accounting closures, so the paper claims the financing contrast as measured sensitivity, not as a theory of money.

Status matrix (Stage 0 answer). Symbols: implemented and validated; partially available – the machinery exists but needs admissibility re-runs, explicit framing, or hardening (the missing piece is stated); not implemented; exists in the code but is inadmissible under the ROADMAP gates (the unfinanced autonomous shock). No cell is yet – under the corrected admissibility standard that is precisely the finding, and the matrix defines the Stage 1 work.

Labour closure	F1: preference reallocation	F2: tax-financed	F3: external debt
BF (friction, η^*)	geometric η implemented; missing η^* pre-registration and the explicit $\tilde{\beta}$ experiment framing	needs the g_i vector and the T rule (the existing unfinanced autonomous shock is as input)	needs the external-account closure
ALPHA (full employment)	normalized composition shock runs (the +0.067% η -invariance diagnostic); needs explicit framing and reporting	the autonomous-demand version is unfinanced; the tax rule is missing	
BETA (elastic supply)	closure not implemented		
GAMMA (sticky real wage)	:fixed implemented; composition run needs a validation pass	the +19.3 pp extensive-margin sweep exists but rests on the unfinanced shock; must be regenerated under F2	
DELTA (IO endpoint)	corner not implemented (fixed factor price + Leontief limit)		

Stage 0 remaining items: (i) audit-trail disclosure (brief mention vs. full appendix vs. none; recommended: brief, per the supersession-note culture); (ii) scope locks: skill-class disaggregation (mandatory only for policy-evaluation claims), open-economy depth beyond the section 4.1 mapping (Armington extension optional), B&F replication off the critical path (confirmed by all surviving plans).

4.1 Intermediate results: the cbase2 pipeline (v3 open-economy recalibration) [added v4]

The notebook pipeline `cbase2/` (see `cbase2/documentation.md` for the neutral description and `cbase2/process_comments.md` for the dated observations) has implemented and verified the v3 open-economy recalibration. Three structural findings shaped it. First, the DELTA investigation showed that the fixed-wage rows are indeterminate without marginal leakages: with the import margin alone, the only solution is the degenerate corner $E = 0$, because the import leak has no offsetting injection. Second, the national-accounts identity $S = I + X - M$ forces the saving rate: the household's non-consumption share is the accounting partner of the exogenous investment and export injections, so the v3 calibration derives it from the data ($s = 0.398$; government share $\tau_0 = 0.214$, export share 0.422, investment share 0.162) instead of assuming it. Third, the equilibrium formulation was corrected throughout: the mobile system keeps N-1 clearing equations plus the $CPI = 1$ numeraire, and the omitted N-th market is the residual external account (it is NOT Walras-redundant once imports leak, so it is exposed and asserted rather than silently dropped; ADR-0010); the F1/F3 tax is the price-indexed baseline budget $T = p'gG$ (F3 reads exactly as "baseline tax unchanged, programme externally financed"), and F2 keeps the genuinely balanced-budget rule.

Verified results (acceptance test `v2_verify.jl`): the injection continuation runs in 5.4 s with no stalls; the price-explosion branch found by the direct solve is eliminated; the F1/F2/F3 mobile rows solve to residuals of at most $3.7e-7$ with EXACT budget identities ($\sum p c = (1-s)E$); the F3 external balance is recorded ($F = 0.00147$ at the $m = 1$ programme); and the DELTA analytic equivalence is EXACT for both F2 and F3 (rel y error 0.0; $L = 1.1656$ and 1.1767 at the $m = 1$ programme). The updated status matrix, same layout as above:

Labour closure	F1: preference reallocation	F2: tax-financed	F3: external debt
BF (endpoint selector, $\eta \in \{0, 1\}$)	machinery verified; the explicit $\tilde{\beta}$ tilt ported to the canonical pipeline ($d = 1 + G0 \cdot \psi / c0$, <code>f1_shift =</code> <code>tilt_g0_over_c0</code>); matrix-cell runs pending	implemented and verified (balanced-budget rule; resid 1.1e-7)	implemented and verified (external balance recorded)
ALPHA (full employment) BETA (elastic supply)	v3 re-run pending ($\eta = 1$) implemented (η_s , continuation solver; elasticity identification verified at the v2 stage); v3 verification pending	v3 re-run pending ($\eta = 1$) verification pending	v3 re-run pending ($\eta = 1$) verification pending
GAMMA (sticky real wage)	v3 run pending	the DELTA corner (its Leontief limit) verified exact; the CES-elasticity GAMMA run pending	v3 run pending
DELTA (IO endpoint, corner)	F1 cell pending	implemented and verified – EXACT analytic equivalence (rel y error 0.0)	implemented and verified – EXACT analytic equivalence (rel y error 0.0)

Open items carried into Stage 1: the BETA verification (the η -continuation runtime under v3), the ALPHA and GAMMA v3 re-runs at $\eta = 1$ and the CES elasticities, the F1 tilted-cell runs (the explicit tilt is ported to the canonical pipeline as `f1_shift = tilt_g0_over_c0`; the matrix cells are pending), and the interpretive pass on the v3 baseline units (the no-numeraire units experiment is retired with ADR-0010: the CPI = 1 numeraire is restored, so the CPI-normalized wage is the real wage; the 0.54 reading belonged to the retired no-numeraire units and must be re-checked in notebook 04).

Post-v4 findings (folded into this version, no version change). Three further results qualify the intermediate state. (i) The branch mystery was resolved: the “depressed” equilibrium found by the solver is the same real allocation at a lower price scale; the fixed nominal tax made its real burden price-level-dependent, so the government budget was made homogeneous (real purchases gG financed at current prices, $T = p'gG$) and the CPI numeraire restored as the scale selector. (ii) Sector 71 was dropped on documented grounds (residual catch-all, 1.8% of gross output, 37.5% self-loop), but the instability then moved to other high-self-loop service sectors – the (f) provenance diagnostic showed the diagonals are genuine data (self-shares identical before and after the import split; 23 sectors above a 0.3 self-share), so no data correction is available and the trade-off is structural. Three dataset variants are now defined in `cbase2/src/calibration.jl` (`full` = 71 sectors, `70s` = drop 71, `reduced` = additionally drop the self-share > 0.45 class 13, 18, 19, 48, 53, 58, 68; coverage: 100%/99.2%/87.4% of gross output). (iii) The solver question is reframed: the theta-homotopy showed that even at $\theta = 2$ (gross substitutes, no self-referencing spiral possible) the FD-Newton stalls at the same 3.6×10^{-4} floor – the difficulty is solver-setup reliability (kinked residuals under ForwardDiff), not model structure; the identical system converged to 2×10^{-10} under a different init. This matches the Baqaee–Farhi practice of solving equilibria as general NLPs (KNITRO/fmincon in MATLAB): the recommended cbase2 route is a constrained-NLP formulation (IPOPT via JuMP, the free KNITRO analogue) if the quick finite-difference/least-squares experiments do not suffice.

The v4 numbers in this section are the CLAMPED cbase2 calibration and are superseded. The A-bill calibration (ADR-0012, ADR-0013) re-anchors the saving rate to $s = 0.1199$ (not 0.398), deletes the clamp entirely (the household residual is the table’s own domestic household column, with zero negative sectors), books the product taxes on intermediate use (raw-table row 75) as the third external leak, and makes the omitted-market canary identity exact at 1.7e-16 on full-71 (9.6e-17 on the 70s variant). The fixed-wage indeterminacy reported above is superseded too: ADR-0014 replaced the heuristic admissibility guard with the verified round-gain criterion and the open calibration is determinate (full-rank Jacobian; three inits converge to one root). The raw table’s own production-vs-expenditure residual (5.387 %) is unrelated and remains open.

4.2 Matrix results: the 5 x 3 evaluation on the A-bill calibration [added v5]

Design `matrix_5x3_v3` (the ADR-0014/ADR-0015 generation): 15 of 15 cells executed, no locked and no dropped scenarios. Below is the pre-registered headline set – Tornqvist real GDP against the baseline, employment, and the model’s own identity-consistent external position (the omitted-market canary, $S - (I+X-M) + T$). The per-cell accounting decomposition

(saving, tax, investment plus exports, and the three import components) is in `paper/tables/matrix_5x3_v3_flows.md`; the paper cites the `matrix_5x3-v3-*` run ids.

Cell	Real GDP rel.	Employment	Net external position
BF-F1	+0.000433	1.000000	+0.000453
BF-F2	-0.016936	1.000000	-0.000548
BF-F3	-0.000000	1.000000	-0.007947
ALPHA-F1	+0.000433	1.000000	+0.000465
ALPHA-F2	-0.016936	1.000000	-0.000601
ALPHA-F3	+0.000000	1.000000	-0.008495
BETA-F1	+0.000433	1.000000	+0.000465
BETA-F2	-0.016936	1.000000	-0.000601
BETA-F3	+0.000000	1.000000	-0.008495
GAMMA-F1	-0.000806	0.999027	-0.000000
GAMMA-F2	-0.015333	1.001260	-0.000000
GAMMA-F3	+0.022644	1.017796	-0.000000
DELTA-F1	-0.000806	0.999027	-0.000000
DELTA-F2	-0.015333	1.001260	-0.000000
DELTA-F3	+0.022644	1.017796	-0.000000

Findings against the pre-registered signatures – deviations are findings, not failures (ADR-0004):

1. The F1 column is admissible in the fixed-wage rows. **GAMMA-F1** and **DELTA-F1** were blocked by a heuristic guard whose premise is false on the open calibration: the Jacobian at the F1 point is full rank (sigma-min/sigma-max = 0.1586) and three different inits converge to the same root. The verified criterion (round-gain column sums below 1) admits them and still rejects closed fixtures, where the sums are exactly 1. The same criterion exposed a latent bug: the retired heuristic had admitted closed-fixture solves whenever manna was present, although manna is a constant and cannot remove a unit root.
2. BETA reproduces ALPHA to machine precision. Under a demand-only programme the price block is demand-invariant ($p = 1$), so the equilibrium real wage sits at its anchor and $L = Lbar * (w/w_0)^{\eta_s} = Lbar$ for any η_s : the supply elasticity is unidentified by THIS experiment, not mis-specified. It bites under a supply-side scenario (a +20 % sector-1 shock moves the real wage to 1.0049 and separates $\eta_s = 0.5$ from $\eta_s = 2$: $L = 1.0024$ versus 1.0098).

Finding 2 is now a measured identification result, not a probe. The supply arm (ADR-0023) is executed: `supply_etas_s1` ($A_1 = 1.1/1.2/1.3$ on sector 1), `supply_etas_prog` ($A_i = 1 + \alpha\psi_i$ on the programme's own sectors, $\alpha = 0.05/0.10/0.20$) and `supply_etas_unif` ($A_i = 1.01/1.02/1.03$ uniform), 63 cells each on the identification matrix ALPHA control + scalar single-wage BETA ($\eta_s = 0.5/1/2/5$) + sectoral uniform BETA ($\eta_s = 0.5/2$), times F1/F2/F3. All 189 cells execute with every gate passing. The implied elasticity $\ln L / \ln w$ recovers the input η_s exactly (0.5000/1.0000/2.0000/5.0000) in each scalar BETA cell and is invariant to the shock magnitude (spread $< 10^{-3}$); the ALPHA control pins $L = 1$ with $w = 1.0049$ (s1) / 1.0046 (prog) / 1.0408 (unif, $A = 1.02$); financing neutrality $F2 = F3$ holds under every supply shock; and the ADR-0019 external-account canary is asserted at $A \neq 1$ on all 189 cells — the gate that had only ever been tested at $p = 1$. The recombination note's (η, η_s) 2 x 2 is thereby realized in its informative regime: the supply arm separates the allocation margin (composition) from the supply elasticity (the level), exactly as the note anticipated. Flow tables: `paper/tables/supply_etas_{s1,prog,unif}_flows.md`; workplan and signatures: `docs/ETAs.md` (v5). 3. The F2 column is not "aggregate 0" for the mobile rows: real GDP falls 1.69 % with employment pinned at exactly 1.0 – a composition effect at fixed aggregate labour input, for a tax withdrawal of 1.33 % of GDP. 4. The limit relations hold: BF at $\eta = 0$ coincides with ALPHA in F1 and F2 at the aggregate, and GAMMA coincides with DELTA to 5e-12 in F2 and F3. F3 is the extensive-margin column – employment rises 1.78 % in the fixed-wage rows. 5. The external account: the F3 programme's gross inflow is $F = 0.0133$ of GDP, of which 0.0029 is its own import content; the NET position is -0.0085 in the mobile rows (the endogenous response offsets 36 % of the inflow) and zero to machine precision in the fixed-wage rows, where all N markets clear and the inflow is absorbed by imports.

Recombination option (left out). The matrix varies two labour margins that the design parametrises independently: BF's allocation selector $\eta \in \{0, 1\}$ (who gets the labour) and BETA's aggregate supply elasticity η_s (how much labour in total). The 5 x 3 samples three of the four corners – ALPHA ($\eta = 1, \eta_s = 0$), BETA (1, 0.5) and BF (0, 0) – and leaves out their recombination, the corner ($\eta = 0, \eta_s = 0.5$): immobile allocation WITH elastic supply. It is left out deliberately because it would be a null result here: with $p = 1$ the real wage sits at its anchor, so the recombined cell coincides with BF to about 1e-15, exactly as BETA coincides with ALPHA. It becomes informative only in the supply-side scenario of

finding 2, where a (η, η_s) 2 x 2 would separate the two margins – the allocation friction moving composition and the supply elasticity moving the level. Recorded as an available extension, not as a gap in the current design.

Version 6 measurement update (ADR-0018). The v5 table above reports the household-consumption Tornqvist index in its "Real GDP rel." column; under ADR-0018 that index is the welfare aggregator, not GDP. Real GDP is the income-side index $w \cdot \text{sum}(L)$ deflated by the seven-component GDP deflator, which is exactly one here because the design pins all prices at one. The matrix_5x3_v4 generation records both concepts; the table below is taken from the 15 v4 manifests. Finding 3 above is corrected: the F2 mobile rows are GDP-neutral (0.000 percent), and the -1.69 percent is the tax-induced welfare loss.

Cell	Real GDP rel.	Consumption rel.	Employment	Net external position
BF-F1	+0.000000	+0.000433	1.000000	+0.000452
BF-F2	+0.000000	-0.016936	1.000000	-0.000548
BF-F3	+0.000000	+0.000000	1.000000	-0.007947
ALPHA-F1	+0.000000	+0.000433	1.000000	+0.000465
ALPHA-F2	+0.000000	-0.016936	1.000000	-0.000601
ALPHA-F3	+0.000000	+0.000000	1.000000	-0.008495
BETA-F1	+0.000000	+0.000433	1.000000	+0.000465
BETA-F2	+0.000000	-0.016936	1.000000	-0.000601
BETA-F3	+0.000000	+0.000000	1.000000	-0.008495
GAMMA-F1	-0.000973	-0.000806	0.999027	-0.000000
GAMMA-F2	+0.001260	-0.015333	1.001260	-0.000000
GAMMA-F3	+0.017796	+0.022644	1.017796	-0.000000
DELTA-F1	-0.000973	-0.000806	0.999027	-0.000000
DELTA-F2	+0.001260	-0.015333	1.001260	-0.000000
DELTA-F3	+0.017796	+0.022644	1.017796	-0.000000

The full per-cell accounting decomposition – the seven GDP components (C, G plus programme, I, X, M_final, M_int, T_int), the external wedge $\text{Sigma}(V) - w \cdot \text{sum}(L)$, and the deflator – is in `paper/tables/matrix_5x3_v4_flows.md`. Findings 1, 2, 4 and 5 of the v5 list stand: the limit relations, the unidentified supply elasticity under demand-only shocks, and the external-account reading are unchanged, and the v4 runs reproduce the v3 equilibria to 1.2e-11, so only the measurement changed. The recombination option remains a deliberate omission.

[added v7: the ADR-0019 generation]

Design `matrix_5x3_v5` (the ADR-0019 generation): 15 of 15 cells executed under the all-N closure with the explicit external transfer, no locked and no dropped scenarios. The headline set below adds the booked external account (F, B_gov, net position F + B_gov) to the v6 columns; the per-cell accounting decomposition is in `paper/tables/matrix_5x3_v5_flows.md`, and the paper cites the `matrix_5x3-v5-*` run ids for the twelve non-BF rows and the `matrix_5x3-v6-*` run ids for the BF row, whose $\eta = 0$ closure changed after this table was written (marker below).

Cell	Real GDP rel.	Consumption rel.	Employment	F	B_gov	Net ext. pos.
BF-F1	+0.000000	+0.000433	1.000000	+0.00000000	0.00000000	+0.00000000
BF-F2	+0.000000	-0.016936	1.000000	+0.00000000	0.00000000	+0.00000000
BF-F3	+0.000000	+0.000000	1.000000	+0.00000000	0.01330991	+0.01330991
ALPHA-F1	+0.000000	+0.001432	1.000000	+0.00078481	0.00000000	+0.00078481
ALPHA-F2	+0.000000	-0.018226	1.000000	-0.00101403	0.00000000	-0.00101403
ALPHA-F3	+0.000000	-0.018226	1.000000	-0.01432394	0.01330991	-0.00101403
BETA-F1	+0.000000	+0.001432	1.000000	+0.00078481	0.00000000	+0.00078481
BETA-F2	+0.000000	-0.018226	1.000000	-0.00101403	0.00000000	-0.00101403
BETA-F3	+0.000000	-0.018226	1.000000	-0.01432394	0.01330991	-0.00101403
GAMMA-F1	-0.000973	-0.000806	0.999027	+0.00000000	0.00000000	+0.00000000
GAMMA-F2	+0.001260	-0.015333	1.001260	+0.00000000	0.00000000	+0.00000000
GAMMA-F3	+0.017796	+0.022644	1.017796	+0.00000000	0.01330991	+0.01330991
DELTA-F1	-0.000973	-0.000806	0.999027	+0.00000000	0.00000000	+0.00000000
DELTA-F2	+0.001260	-0.015333	1.001260	+0.00000000	0.00000000	+0.00000000
DELTA-F3	+0.017796	+0.022644	1.017796	+0.00000000	0.01330991	+0.01330991

Marker (same date, within this version): the three BF rows of the table above are the SUPERSEDED pin values – the $F = 0$ pin of ADR-0019 D2 leaves the external position unidentified, so the entry collapses to the programme cost $B_gov = 1.331$ percent of GDP and the account stands open by -0.794 percent at BF-F3. The current BF results (sectoral wages at $\eta = 0$, F solved, identified external position, closed account) are in the promotion block below and in `paper/tables/matrix_5x3_v6_flows.md`; the twelve non-BF rows of this table, GAMMA and DELTA included, are unchanged in v6.

Three earlier readings are superseded by ADR-0019. First, the pre-registered external deficit = F signature: F3's gross programme inflow is $B_gov = 1.331$ percent of GDP while the net position is $F + B_gov$ (-0.101 percent in the mobile rows, $+1.331$ percent in the fixed-wage rows, and not identified at the BF $\eta = 0$ rows; see the correction below). Second, the wedge-as-external-position reading: `gdp_wedge` is now minus the identity gap (about $1e-16$ at the mobile $\eta = 1$ cells, about $1e-12$ at the fixed-wage $\eta = 1$ cells at residual level, and the documented factor-market gap of $+4.34e-04 / -5.62e-04 / -7.94e-03$ at the BF $\eta = 0$ cells), and the external position is the booked $F + B_gov$. Third, the F2/F3 separation: the two columns are real-neutral in the mobile regime (identical consumption at -1.823 percent and identical net position to about $2.6e-17$, with $F_F3 = F_F2 - B_gov$), so their v1-v4 differences were artifacts of the omitted-market shortcut. Findings 1, 2 and 4 of the v5 list stand unchanged, finding 3 stands in its v6 corrected form, and the recombination option remains a deliberate omission.

Correction (same date, within this version): the BF $\eta = 0$ external-position entries are booked financing, not positions. The $\eta = 0$ endpoint pins $F = 0$ (ADR-0019 D2), so the reported entry $F + B_gov$ collapses to the programme cost $B_gov = \text{sum}(p\ g) = 1.331$ percent of GDP – identical to the last digit in BF-F3, GAMMA-F3 and DELTA-F3, i.e. a design constant that no labour closure can move (GAMMA/DELTA-F3 close their account to about $1e-12$, BF-F3 does not). The same cell's resource side $S + T_int + M - (I+X)$ reads $+0.5374$ percent of GDP at BF-F3 against the booked $+1.3310$ percent, the account open by -0.7936 percent of GDP; that gap is exactly the labour-market residual the pin leaves, $-w(\text{sum } L_cm - \text{sum } L)$, with $\text{sum } L_cm = 1.0079361$ against the frozen bar 1 at BF-F3 (verified at all 15 cells; at the twelve $\eta = 1$ cells the two quantities agree to machine precision). The position is therefore not identified at $\eta = 0$: replacing the pin by $F = c$ leaves every c an exact root (residual $2.2e-15$) and moves the entry one-for-one, from -0.669 to $+3.331$ percent of GDP for c in -0.02 to $+0.02$, with the resource side moving too. Solving the $\eta = 0$ system with the pin at ALPHA's solved F reproduces the ALPHA cell to $2.8e-17$ in the canonical vector and in the consumption block, and replacing the pin by the labour equation $\text{sum } L_cm = Lbar$ closes the account to $4e-16$ and returns BF-F1/F2/F3 = ALPHA-F1/F2/F3 exactly (with financing neutrality then holding at $\eta = 0$ as well), so in this demand-only design the BF/ALPHA difference – including the BF-F3 welfare entry of 0.000 percent, which becomes -1.8226 percent at the labour-clearing F – is carried by the pin, not by the frozen allocation. Read the BF rows from the resource side and the gap in `paper/tables/matrix_5x3_v5_flows.md`; the $\eta = 0$ closure options and their scope for a v6 generation are in ADR-0020 (accepted: option C, with the specification and probe measurement at the end of this section).

Sources of the recorded deviations (classification). None of the recorded deviations was an accounting error: the identities (zero-profit, budgets, the booked external account) hold to machine precision throughout. They fall into four classes. (i) Measurement–definition: the legacy "Real GDP" column is the household-consumption welfare index; corrected by the income-side measure (ADR-0018, version 6). (ii) Closure artifact: the F2/F3 separation and the earlier external-position readings came from the omitted N -th market; corrected by all- N clearing with the booked external account (ADR-0019, this version; the approaches considered are catalogued in its 2026-09-18 amendment), under which the mobile F2 and F3 cells are real-neutral. (iii) Model-class property: BETA coincides with ALPHA under demand-only shocks because the one-factor CRS price block is demand-invariant (Samuelson nonsubstitution), and DELTA's exactness rests on the same $p = 1$; η_s is identifiable only under a supply-side scenario or with a second production factor – a property of the closure design, not a mis-specification. (iv) Corrected admissibility mathematics: the GAMMA/DELTA-F1 rejections were a retired heuristic guard, replaced by the measured actual-matrix criterion (ADR-0014, ADR-0017).

Decision (same date, within this version): ADR-0020 is accepted with option C – sectoral wages at the $\eta = 0$ endpoint – so the BF row's external position is repaired in the closure rather than patched by convention. The frozen sectoral allocation stays in place and each sector's wage is set by that sector's own marginal product at the frozen allocation: unknowns $[p; y; w(1:N); F]$, $3N + 1$ in all, with the N zero-profit conditions $p_i = \text{cost_i}(p, w_i)$, the N sectoral first-order conditions at the frozen allocation, the N clearing equations with household wage income the sum of $w_i L_i$, and the CPI numeraire. F is kept because the block of zero-profit, FOC and clearing conditions is homogeneous of degree 1 in (p, w, F) : N sectoral conditions replace the mobile system's single aggregate labour equation, so the block carries one equation more than it has directions to pin and the demand side needs one free scalar; dropping a clearing equation is the ADR-0010 shortcut that ADR-0019 retired. Two consequences, both measured: the external account closes at $\eta = 0$, because the identity gap is exactly the value-weighted difference between the cost-minimizing and the frozen labour allocation and that difference is zero when the frozen allocation is cost-minimizing at these wages; and financing neutrality holds at $\eta = 0$ as well ($F_F3 = F_F2 - B_gov$, with an identical booked position). Option B (labour equation instead of the pin, common wage) is recorded in ADR-0020 as a future venue: it also closes the account, but it collapses the BF row onto

ALPHA at 2.8e-17, so it buys accounting coherence by giving up the immobile benchmark as a distinct economy; option A is superseded by this choice.

Probe measurement (full-71 A-bill calibration, before any src change; `experiments/probes/probe7_sectoral_wages_eta0.jl`):

Cell	max abs residual	identity gap	F	B_gov	booked F + B_gov
BF-F1	1.4e-13	+1.9e-14	-0.00581497	0.00000000	-0.00581497
BF-F2	4.2e-13	-4.2e-13	-0.00852841	0.00000000	-0.00852841
BF-F3	1.7e-12	-5.0e-13	-0.02337469	+0.01484628	-0.00852841

The eta = 0 row is therefore no longer a normalisation of ALPHA: it carries its own wages (0.970 to 1.672 around the CPI-pinned level, wage bill 1.00702), its own prices and an identified external position (F-column norm 0.163; Jacobian condition number 9.4e7 against 52.8 for the mobile all-N system, so the v6 acceptance gate and the polish target must be set from measurement rather than copied). Because the F3 booking is priced at the eta = 0 prices, which the sectoral wages move, it reads +1.4846 percent of GDP there against the baseline +1.3310 percent. Scope of the change: it is the eta = 0 branch only. ALPHA and BETA are the eta = 1 mobile system and GAMMA and DELTA are the fixed-wage regime (w = 1 pinned, no labour equation, no F) – a different system from `problem` – so those rows' v5 results are unaffected by design; and the one shared helper the change touches (`_cost_minimizing_labor`, whose scalar-wage form is $\log(w)$) becomes generic through $\log(w)$, a no-op at a scalar wage, which the promotion must verify by re-running the v5 cells and asserting the twelve non-BF rows reproduce to machine precision. Promotion into `src/` and the `matrix_5x3_v6` generation is the next step and is not yet done.

Promotion (same date, within this version): option C is in the kernel and executed as the `matrix_5x3_v6` generation (15 of 15 cells executed, commit 6db1da5). The change is dispatch-only: `_cost_minimizing_labor` evaluates $\log(w)$ and so accepts a wage vector, `_wage_bill` keeps the scalar arithmetic bit-identical while summing $w_i L_i$ for a vector, `problem_sectoral` is the new eta = 0 system (3N + 1 unknowns), `problem` rejects eta = 0, and `solve` dispatches the new system with a tighter primary tolerance (1e-8) and a longer polish ladder (6 steps, target 1e-13) because it is stiff. Verification: the twelve non-BF cells reproduce their v5 manifests – ALPHA/BETA to 6.7e-16 and GAMMA/DELTA to 1.6e-11, where the same differences appear on the pristine kernel (stash test, `probe9_nonbf_reproduction.jl`), so they are the fixed-wage sensitivity to the warm start (the system is well conditioned: Jacobian cond about 6.3 measured on the GAMMA-F2 cell, so the amplification is in the measurement layer, 100x the residual level) rather than an effect of the change. The eta = 0 identity gate is set from the measured floor (harness 1e-9, test 1e-10; worst 1.04e-11 at BF-F1) instead of the 1e-12 written in ADR-0020's plan text. The BF row now reads:

Cell	F	B_gov	Net ext. pos.	Gap	Consumption rel.
BF-F1	-0.005815	0	-0.005815	-1.0e-11	+0.096 percent
BF-F2	-0.008528	0	-0.008528	-1.8e-13	-1.980 percent
BF-F3	-0.023375	+0.014846	-0.008528	-1.9e-13	-1.980 percent

against the v5 pin generation (net external position 0 / 0 / +1.331 percent of GDP, gap +4.3e-04 / -5.6e-04 / -7.9e-03, consumption rel. +0.043 / -1.694 / 0.000 percent). Financing neutrality now holds at eta = 0 as well (F_F3 = F_F2 - B_gov to 3.5e-14, with identical positions), the BF/ALPHA coincidence is gone, and the paper-facing table is `paper/tables/matrix_5x3_v6_flows.md`. The correction above describes the superseded v5 generation: cite the `matrix_5x3-v6-*` run ids. A harness episode belongs to the record: the first v6 batch recorded the three BF cells as failed on `MethodError: isfinite(::Vector{Float64})`, because the `wage` metric carried the sectoral wage vector at eta = 0; the metric now reports the wage-bill-weighted aggregate (dispersion in the new `wage_min` / `wage_max` diagnostics), the three crash directories were removed and the whole generation re-run so that all fifteen cells come from one harness version. The eta = 0 FOC row also had a real defect before that: using the frozen allocation in the FOC made those N equations identically zero and the system degenerate, which the external-account gate caught immediately (gap 3.3e-04) – the acceptance test earned its keep.

The same generation's fixed-wage rows are unchanged, so the current GAMMA and DELTA results remain those of the v7 headline table above: the promotion was verified row by row (ALPHA/BETA reproduce v5 to 6.7e-16, GAMMA/DELTA to 1.6e-11, the fixed-wage warm-start sensitivity recorded below). For reference, the current GAMMA row is:

Cell	Real GDP rel.	Consumption rel.	Employment	F	B_gov	Net ext. pos.
GAMMA-F1	-0.000973	-0.000806	0.999027	0.00000000	0.00000000	0.00000000
GAMMA-F2	+0.001260	-0.015333	1.001260	0.00000000	0.00000000	0.00000000
GAMMA-F3	+0.017796	+0.022644	1.017796	0.00000000	0.01330991	+0.01330991

These are the values to cite for the fixed-wage rows. DELTA reproduces them to six digits (its Leontief corner), the external account closes to about 1e-12, i.e. at the level of the solver residual, and F3 is the extensive-margin column: employment rises 1.78 percent and no external transfer is involved (the fixed-wage regime has no external unknown).

[added v7, same date: variation in GAMMA]

The fixed-wage rows carry one degree of freedom in the matrix. F2 and F3 are real-neutral by theorem (financing neutrality) and DELTA reproduces GAMMA to six digits, so the three GAMMA cells are effectively two economies; and the flatness is a property of the design rather than of the model class, because the closure's parameter space already contains a wage vector that the design sets to its degenerate value (all sectors equal) and the kernel hard-codes as the scalar pin. ADR-0021 (proposed, same date) would make that vector a parameter of the existing closure, with the wage structure as a preregistered design dimension instead of a new closure; the options, their costs and the claim-by-claim provenance are in `docs/VariationinGamma.md`.

Measured on the GAMMA-F2 cell (full-71 A-bill, `experiments/probes/probe11_gamma_wage_structure.jl`, 2026-09-18; a prototype that mirrors the fixed-wage residual, so the magnitudes await a kernel run): the degenerate pin reproduces the cell (residual 8.9e-16, employment 1.00125981). A UNIFORM rescale of the pin is a numeraire change and moves nothing real – $wbar = 0.9$ and 1.1 leave the allocation, employment and real income invariant to 8.9e-16, with the CPI scaling exactly – so the wage LEVEL is not an instrument. A RELATIVE change is a genuine shock: a ten percent wage push in the seven programme sectors moves employment by about half a percent and real income by about one percent (1.00686452 and 1.01186464 against 1.00125981), a single-sector tilt moves the allocation by about two percent in log terms, and a broad tilt of sectors 1 to 5 by about 3.5 percent – larger than the closure's entire existing F1 to F2 spread. The external account still closes at the solver-residual level under every tilt (price-weighted clearing residual 2.9e-16 at the degenerate pin, at most 7.0e-12 tilted), validated against the kernel canary along quantity perturbations (ratio 1.0000 at one and ten per mille). The Jacobian condition number stays between 6.32 and 6.34, so the fixed-wage system is well conditioned at this calibration and its metric sensitivity to the warm start is a measurement-layer amplification of the residual level rather than a singular solve.

Three limits belong with these numbers. First, the pinned vector is an exogenous scenario parameter, never an equilibrium object: the variation is an instrument response, like the programme incidence or the shock size, and the scenarios must be preregistered before any run that reports them. Second, the level invariance is closure-conditional – it holds because the model has no nominal anchor besides the wage, and it would fail with a nominal government bill, nominal debt or money. Third, the magnitudes come from a mirrored residual: the ADR-0020 promotion showed that translation into the kernel is where defects appear, so they should be reproduced through the kernel before citation, which also requires extending `external_balance_canary` and `gdp_components` so that the acceptance gate evaluates a tilted pin on kernel arithmetic rather than on a proxy. The zero-cost alternative, needing no kernel change, is a sectoral supply-shock design, which also makes the allocation margin bite. The two routes it leaves open are recorded the same date: the thematic supply-side version (a climate-change scenario with a sectoral incidence vector) as `docs/ideas/IDEA-0001-climate-productivity-shocks.md` in the new ideas register, and the demand-side route (a free wage in a segmented wage structure, which is what makes prices demand-sensitive) as `docs/WORKPLAN_SENSITIVE_PRICES.md`. Measured while scoping the latter: the $\eta = 0$ row is already the only row with demand-sensitive prices (max $\text{abs}(p - 1)$ of 0.221 to 0.279 against 2.7e-15 in the ALPHA and BETA rows), because its sectoral wages are solved given demand, while a pinned wage vector produces price heterogeneity without price sensitivity.

5 Workplan [section 5 since v4; reworked v3]

5.1 Stage 1: Model completion [reworked v3]

Ordered by the status matrix – the financing core first (it makes every experiment admissible), then the missing labour closures, then the guards:

1. **Financing core:** the shared final-demand and institutional-budget core; the government/investment vector g_i with the tax rule $\sum_i p_i g_i = T$ (F2); the external-account closure with $\sum_i p_i g_i = F$ and import content at fixed

sector shares (F3); the explicit preference-reallocation experiment with renormalized weights $\tilde{\beta}_i$ (F1); retire the unfinanced autonomous shock (). – DONE in v3, superseded in scope: the v3 open-economy calibration implements the government block, investment, exports, import margins, the saving rate and all three financings (see section 4, intermediate results). The explicit $\tilde{\beta}$ tilt is ported to the canonical pipeline (`experiments/run.jl, f1_shift = tilt_g0_over_c0`); the tilted matrix cells are pending.

2. **BETA**: elastic total labour supply on the real wage, $L^s = \bar{L} [(w/P)/(w_0/P_0)]^{\eta_s}$, with the labour–leisure interpretation and the numerical elasticity test $d \log L / d \log (w/P) \approx \eta_s$. – Implemented (η_s , η -continuation solver, single-point elasticity identification verified at the v2 stage: 0.5/1.0/2.0 recovered); the v3 verification run is pending (η -continuation runtime).
3. **DELTA corner**: fixed real factor price, unconstrained factor quantity, Leontief limit of the CES core; document the exact equivalence conditions (R2’s own point, answered). – DONE: implemented as `:fixed + Leontief limit`; the analytic equivalence is EXACT for F2 and F3 (rel y error 0.0).
4. **Pre-register the BF endpoints** before any Stage 2 run: only $\eta \in \{0, 1\}$ are kept (ADR-0010); no interpolated η^* .
5. **Cobb-Douglas limit guard** (Milestone C): sign-safe real powers at $\varepsilon \rightarrow 1$. – Implemented (analytic branch at $\varepsilon = 1$); the continuous-epsilon assertion is pending its v3 re-run.
6. **Full residual validation** (ROADMAP Phase 4): **all-N clearing** (ADR-0019; the omitted-market rotation is retired), household-expenditure exhaustion for every experiment type, homogeneity, multi-start convergence, Tornqvist consistency. Machine-precision residuals are the gate. – Partially done: budget identities exact at every solved equilibrium; the fixed-wage regime enforces all N clearings, the mobile regime enforces N-1 plus the CPI numeraire and asserts the omitted N-th market against S - (I+X-M) (ADR-0010); the omitted-equation rotation and multi-start battery are pending. – v7 update: every regime now enforces all N clearings with the explicit external account (ADR-0019); the mobile $\eta = 1$ identity $S + T_{\text{int}} + M - (I+X) = F + B_{\text{gov}}$ and the all-N clearing residuals are gated in tests and experiments, with the $\eta = 0$ factor-market gap reported rather than gated. The equation-ordering rotation is retired with the omitted market; a multi-start battery on the full grid remains open.

5.2 Stage 2: Simulation and sensitivity [reworked v3]

1. **Fill the 5 x 3 matrix**: headline set per cell (Tornqvist real GDP, employment, external balance), sectoral detail into the appendix; compare each cell against its pre-registered expected signature – deviations are findings, not failures. – DONE (v5): all 15 cells executed on the A-bill calibration (section 4.2); the F1 column’s fixed-wage cells became admissible once the heuristic guard was replaced (ADR-0014). The recombination option (allocation x supply elasticity) is recorded in section 4.2 as an available extension, deliberately left out because it is a null result for a demand-only programme.
2. **Sectoral Sobol**: does the reallocation friction (BF) matter for sectoral allocation even where it is aggregate-second-order? (R1.7’s aggregate-vs-sectoral demand.)
3. **Robustness**: shock magnitudes; the financing variants are already matrix columns; state the B -collapses-into- T equivalence as a limitation.
4. **Elasticity table** (R1.3): values, ranges, sources, distributions, pre-registered.

Exit criterion (ROADMAP Phase 5): headline rankings qualitatively stable across financing closures and shock sizes; otherwise ranking instability itself is the result.

5.3 Stage 3: Manuscript and reviewer response [reworked v3]

The manuscript is written from the two-wedge design; every referee point is addressed at a named location. The nine-section structure (definitive guide Part III, adapted):

1. **Introduction** – the two wedges; state upfront that the IO–CGE bridge is known (Robinson 2006, Rose 1995); one short Kuhnian paragraph at most (R2.3; R1.1).
2. **Related literature** – the closure taxonomy; Robinson (2006) central; Rose (1995); Dervis–de Melo–Robinson (1982); de Melo & Tarr (1992); McGregor–Swales–Yin (1996); Shoven & Whalley (1984); Mansur & Whalley (1984); Willenbockel (1994) (R2.2; R1.8).
3. **Data and accounting** – the section 4.1 pipeline, import separation, decomposed value added, the documented 5.387% residual, the calibration table (R1.3; R2.4).
4. **Model and closures** – verbal model overview before equations (R1.2); static horizon stated; the two wedges as design dimensions; financing closures explicit (R2.4, R2.5).
5. **Aggregate results** – matrix rows; wage-regime vs elasticity channels (R1.7).
6. **Sectoral results** – sectoral Sobol, bottlenecks (R1.7).
7. **The financing wedge** – bracket results, the external-debt variant, the bounded endogenous-money paragraph.

8. **Robustness and limitations** – financing variants, shock sizes, one-factor aggregation, static horizon, open-economy treatment.
9. **Conclusion** – practical closure guidance; no bridge-discovery claims (R2.1/R2.3 answered by measurement).

Reviewer 1 point map: R1.1 structure/central argument -> sections 1 and 3 (the matrix); R1.2 verbal model description -> section 4; R1.3 calibration summary table -> section 3 and the elasticity table; R1.4 Type-I/II multipliers -> clarify or drop in section 2; R1.5 wage-setting rule -> GAMMA/BETA with explicit floors; R1.6 mobility/unified wage -> BF/ALPHA; R1.7 aggregate vs sectoral -> sections 5–6 with the sectoral Sobol; R1.8 McGregor–Swales–Yin -> literature and the DELTA corner; R1.9 minor copy-edits (p. 5 sentence, p. 16 paragraph, Figure 3 axis).

Reviewer 2 point map: R2.1 labour-supply elasticity as bridging parameter -> BETA and the matrix; R2.2 closure literature -> section 2; R2.3 cleavage framing -> domain-of-applicability reframing, section 1; R2.4 closed-economy miracle -> section 3 pipeline plus the external-debt closure; R2.5 sector-specific immobile labour -> the mobile core (ALPHA/BF); R2.6 eq. (19) sign error -> section 4 fix; R2.7 supply-vs-demand characterization -> removed; R2.8 CES novelty claim -> dropped; R2.9 Cobb-Douglas archetype -> dropped; R2.10 arbitrary uniform allocation -> eliminated by endogenous allocation (BF/ALPHA); R2.11 "fundamentally contested" allocation claim -> dropped or referenced; R2.12 internal mobility contradiction -> resolved by the mobile core; R2.13 McGregor et al. -> section 2 and DELTA; R2.14 skill classes -> optional extension with the limitation stated.

Methodological correction: the corrected-equilibrium story (Result 1 of the definitive guide) is the response letter's opening; the disclosure level remains a Stage 0 decision. Must not appear: all η -dominance claims, the 88.4%/100% figures, GO certification language, price-invariance claims, and "the IO multiplier requires $\eta \rightarrow \infty$ " (it requires sticky wages).

5.4 Stage 4: Response letter and submission [reworked v3]

Reviewer 2: maximally gracious; concede "the bridge exists" fully; present the corrected equilibrium and the measured matrix as the contribution. Reviewer 1: point-by-point via the map above; calibration table; McGregor engagement; aggregate-vs-sectoral quantification. Then the final gates (ROADMAP section 11): clean-checkout reproduction, reconciled accounts, explicit financing, identity tests, sensitivity shares conditional on stated distributions, tag the submission commit.

5.5 Immediate next step [reworked v3]

Stage 0 is resolved except the disclosure decision and the scope locks. Begin Stage 1 with the financing core (g_i , the T rule, the external account, the explicit F1 experiment), then BETA, then the DELTA corner; pre-register the BF endpoints ($\eta \in \{0, 1\}$, ADR-0010) before any Stage 2 run.

Update (v4): the financing core and the DELTA corner are done; the remaining Stage 1 items are the BETA verification run, the ALPHA/GAMMA v3 re-runs ($\eta = 1$, CES elasticities), the F1 tilted-cell runs (tilt ported to the canonical pipeline; cells pending), and the interpretive pass on the v3 baseline units in notebook 04 – then re-preregister the design ($\eta \in \{0, 1\}$, ADR-0010) and open Stage 2.

5.6 Running the pipeline (container and Mac) [added v4, folded]

Execution venue. Notebooks 01 (data wrangling) and 02 (accounting consistency) are validated end-to-end and reproduce the parent AC_* artifacts exactly; notebook 03 (financing closures) needs a v3 re-run once the solver route is settled. Notebooks are executed in the container (Julia 1.12.7, depot /opt/julia-depot, kernel julia-1.12); the Mac is a second venue for the same code, useful for solver experiments, not for speed.

Can runtime be ruled out? Yes, in the narrow sense: a single 140-dimensional solve takes seconds and the failures are convergence stalls at floors that are identical across algorithms and run-times – more CPU time does not fix them. A Mac run is nevertheless NOT superfluous, but its purpose is solver robustness in a clean environment: the session evidence is init/path-sensitive (the identical system converged to 2×10^{-10} under one init and stalled at 3.6×10^{-4} under another), so a controlled init sweep is the cheapest decisive experiment.

Commands. Container (repo root BFRP/(3)BeyondHulten): `julia --project=. -e 'using Pkg; Pkg.instantiate()' once; acceptance test julia --threads=4 cbase2/scripts/verify_v3.jl (the standing gate: baseline continuation, F1/F2/F3 rows, budget identities, DELTA equivalence). Mac (same repo checked out, juliaup 1.12.x): first julia --project=. -e 'using Pkg; Pkg.instantiate(); using BeyondHulten'; then (1) the acceptance test as above, (2) an init sweep on one calibrated economy – default [ones(N); lambda; 1.0], warm linear fixed point, and small random perturbations of each – over $\theta \in \{2.0, 1.0, 0.5\}$ with drops=[71], recording retcode and residual per run (a short script to be added as cbase2/scripts/solver_sweep.jl); (3) the same with AutoFiniteDiff() and with Levenberg–Marquardt`

as primary. If any route reaches machine-tolerance residuals, the pipeline proceeds unchanged; if all stall, implement the IPOPT/JuMP formulation.

The chronological document map, the per-document evaluation of documents 1–10, and the synthesis have been moved to `docs/archive/document_review_leftovers.md` at Version 3; this working document now contains the foundations, the evaluation matrix, and the workplan only.

6 Revision Log [added v2]

- **Version 1** (September 2026) — Initial chronological assessment of `docs/`.
- **Version 2** (September 2026) — Fixed a LaTeX build error (backticked paths with underscores inside `\texttt{\textcolor{span}}` spans); added the chronological roadmap chapter (Stages 0–4) with Decision overview I (labour-market closures) and Decision overview II (financing closures) as Stage 0 tables with recommendations, folding in all still-relevant details from documents 1–8; documents 1–8 moved to `docs/archive/` (`definitive_guide.md` and `labor_closures.md` remain active); version history block added; roadmap restructured into Foundation I (labour-market closure) and Foundation II (financial closures) with per-option two-column provenance tables (formulation, status, referee point, cost, recommended role, reviewer, old docs, Baqaee–Farhi usage), followed by the workplan; corrected the geometric reallocation row (“used by BF” – the BF 2019 replication ships both the immobile and the mobile variant, verified in the original MATLAB files); added the exogenous labour-endowment shift as a ninth closure table (BF 2022 BLS labour-supply shifts; the original paper’s slack injection); extended the coverage check with firm capacity constraints and adjustment-cost frictions, plus a to-verify nuance on the BF 2019 mobile variant.; added a lineage note under the exogenous-shift table documenting the direct resonance with the original manuscript’s Section 5 (slack narrative borrowed from BF 2022’s social-distancing channel; code mapping of the three slack implementations to the manuscript’s two variants); Foundation I regrouped into the mandatory minimum (reallocation margin, Closure A, sticky wage, Closure D) and the extended option set (Closure B, partial mobility, Closure C, legacy callback, exogenous shift); final selection implemented: 1.1 selected closures BF (explicitly justified vis-a-vis BF 2019’s neoclassical short-run market-friction argument) and ALPHA–DELTA (neoclassical standard, neoclassical labour–leisure, production-agnostic Keynesian, Leontief-inverse emulation) with a pinned suggested narrative; 1.2 notable unselected candidates (partial mobility, complementarity, exogenous shift as scenario device); 1.3 dropped legacy callback; financing deduction routes clarified (household composition vs. government budget vs. lump-sum tax), static-horizon nuance added (debt collapses into the tax variant; foreign financing is the genuinely distinct robustness), recommendation updated to tax principal with preference-reallocation and foreign robustness; header numbering repaired (subsection level under the foundation sections), coverage check moved to Foundation I as its fourth subsection, plausibility parenthetical removed; financing design set per user decision: principal bracket of preference reallocation and tax financing, external-debt (foreign financing) third variant as the static analog of the endogenous-money intuition, imports absorbed mechanically at fixed sector import shares (Armington extension flagged via the existing Omega split); new section “The Evaluation Matrix (5 x 3)” added with pre-registered expected signatures per cell, the limit relations (BF endpoints, BETA to GAMMA, GAMMA plus Leontief = DELTA corner), and the sketch-derived figure “Paradigmatic directions towards the short run” (`docs/pictures/paradigmatic_directions.png`); Foundation I merged the exogenous-shift table into the Dropped section (one closure, legacy callback = endowment shift); Foundation II restructured into Selected (bracket + external debt) and Dropped (expenditure switching; pure debt subsumed by the tax variant) with a final endogenous-money paragraph; `compile-all.zsh` repaired (pandoc now called with `-resource-path` so relative image paths resolve against the `md` directory).
- **Version 3** (September 2026) — Stage 0 answered by the status matrix (5 x 3 with symbols: what exists, what is missing per cell); the second paradigmatic wedge named (financing; loanable-funds vs endogenous-money) with a bounded-strength recommendation; Workplan reworked (Stages 0–4) with inline reviewer-response maps (R1.1–R1.9, R2.1–R2.14) and the two-wedge manuscript structure; chronological review sections moved to `docs/archive/document_review_leftovers.md`.
- **Version 4** (September 2026) — Intermediate implementation results of the `cbase2/` pipeline reported: the Stage 0 chapter promoted to its own section (section 4) with an updated status matrix; the remaining stages form the Workplan section (section 5). New subsection “Intermediate results” records the three structural findings of the v3 open-economy recalibration (the fixed-wage indeterminacy without leakages and the degenerate $E = 0$ corner; the national-accounts identity $S = I + X - M$ forcing the saving rate $s = 0.398$, calibrated from the data together with $\tau_0 = 0.214$, export share 0.422, investment share 0.162; the corrected equilibrium formulation with N-1 mobile clearings plus the CPI numeraire and the asserted residual external account, the price-indexed F1/F3 tax and the balanced-budget F2 rule) and the verified results (injection continuation 5.4 s without stalls, price-explosion branch eliminated, F1/F2/F3 mobile rows at $\text{resid} \leq 3.7e-7$ with exact budget identities, F3 external balance recorded,

DELTA analytic equivalence EXACT for F2 and F3). Stage 1 items annotated with their state (financing core and DELTA corner done; BETA implemented with the verification run pending; CD guard implemented; residual validation partially done); immediate next step updated (BETA verification, ALPHA/GAMMA v3 re-runs, F1 tilted-cell runs, notebook 04 interpretive pass, then pre-registration).

- **Version 5 (September 2026)** — The executed 5 x 3 matrix and the records that made it runnable. New section 4.2 reports all 15 cells of the `matrix_5x3_v3` design (headline set plus the findings against the pre-registered signatures) and records the recombination option (the allocation x supply-elasticity corner) as a deliberate omission. Section 4.1's clamped-calibration numbers and its fixed-wage indeterminacy finding are corrected in place (ADR-0012/0013/0014); section 3's signature table gains the same pointer; Stage 2's matrix item is marked done. Accounting: the A-bill calibration (row 73 domestic bill, row 74 imports, row 75 product taxes as three booked leaks) makes the external-identity canary exact at $1.7e-16$; the raw table's 5.387 % production-vs-expenditure residual stays open. Solver: the chase2 ladder is retired (DE-0010) and the polish target is $1e-10$ (ADR-0015), so cell metrics no longer depend on where the solver stopped. One pre-existing unescaped percent sign (Stage 3, the documented production-vs-expenditure residual) is escaped so the line no longer truncates its tail; no other text changed.
- **Version 6 (September 2026)** — Real GDP and welfare separated in the measurement layer (ADR-0018): income-side real GDP *wsum(L) deflated by a seven-component Tornqvist GDP deflator, the expenditure Divisia index as its dual, and the external wedge Sigma(V) - wsum(L)* reported as a diagnostic. New kernel API (`gdp_components`, `gdp_deflator`, `gdp_income`, `gdp_expenditure`, `gdp_wedge`, `real_consumption`); manifest schema v2 records the gdp and consumption metric keys plus the seven component diagnostics. Because the measurement layer changed, the matrix is re-minted as `matrix_5x3_v4`: 15 of 15 cells executed, all gates pass, and the v4 equilibria reproduce v3 to $1.2e-11$. Section 4.2 carries the v6 table and corrects the v5 F2 reading (mobile income GDP flat, -1.69 percent is welfare); the manifest-generated flow tables are in `paper/tables/matrix_5x3_v4_flows.md`. The F2 open gate in `registry/closures.toml` is resolved by the same runs.
- **Version 7 (September 2026)** — The external account harmonized (ADR-0019): all N clearings enforced in every regime, the mobile $\eta = 1$ system carrying the endogenous net external transfer F and F3 booking the programme financing as `B_gov`, with the booked identity $S + T_{int} + M - (I+X) = F + B_{gov}$. The matrix is re-minted as `matrix_5x3_v5` (15 of 15 cells executed; v1-v4 stay as history); section 4.2 carries the v7 headline table with the booked external account and corrects the superseded readings (external deficit = F, the wedge-as-external-position, the F2/F3 separation); the manifest-generated flow tables are in `paper/tables/matrix_5x3_v5_flows.md`. The generation's finding is financing neutrality (F2 and F3 real-neutral in the mobile regime; the v1-v4 F2/F3 differences were omitted-market shortcut artifacts). Measured movement against v4: mobile F1 welfare +0.043 percent to +0.143 percent, mobile F2/F3 welfare -1.694 percent to -1.823 percent, fixed and BF rows unchanged in their real allocation; the BF $\eta = 0$ external-position entry moves by definition, from -0.7947 percent (the v4 residual canary) to +1.3310 percent (the booked `B_gov`), and is corrected in section 4.2 of this version as booked financing over an open account rather than an identified position (ADR-0020, accepted the same day with option C: the $\eta = 0$ endpoint moves to sectoral wages, the account then closes and financing neutrality extends to $\eta = 0$; the specification and the probe measurement are at the end of section 4.2). **Option C was promoted and executed the same day: the matrix is re-minted as `matrix_5x3_v6` (15 of 15 cells, commit 6db1da5), in which the BF row carries an identified external position (-0.581 / -0.853 / -0.853 percent of GDP) with the account closed to $1.0e-11$ or better and financing neutrality extended to $\eta = 0$, while the twelve non-BF cells reproduce v5 (ALPHA/BETA to $6.7e-16$; GAMMA/DELTA to $1.6e-11$, the fixed-wage sensitivity to the warm start (the system is well conditioned: Jacobian cond about 6.3 measured on the GAMMA-F2 cell, so the amplification is in the measurement layer, 100x the residual level), verified to be independent of the change). `matrix_5x3_v5` and the correction above stay as the record of the superseded pin. The same date added, within this version: the current GAMMA row as an explicit reference block (unchanged from the v7 headline table, since the promotion reproduced the twelve non-BF cells), and a note on variation in GAMMA — the one-degree-of-freedom diagnosis, the measured wage level invariance and wage structure responses, the closed external account under a tilted pin, the conditioning result, the three limits that belong with the numbers, and the pointer to `docs/VariationinGamma.md` and ADR-0021 (proposed). The F3 definition gates in `registry/closures.toml` are resolved by the same runs; the `revisionV6` colour used here assumes the render template defines it (as `revisionV5` was introduced for Version 6). Section 4.2 also classifies the sources of the recorded deviations (measurement definition, closure artifact, model-class property, corrected admissibility mathematics), and the ADR-0019 amendment of the same date records the harmonization approaches considered (status-quo canary, all-N plus transfer F, savings-driven investment, Armington expenditure switching, direct external-balance imposition).**
- **Version 8 (September 2026)** — The supply arm is executed (ADR-0023): 189 cells across `supply_etas_s1/prog/unif`, all gates pass. η_s is now identified under supply shocks (implied $\ln L / \ln w = \text{input to } 1e-3$, magnitude-invariant); the ALPHA control pins $L = 1$; F2 = F3 financing neutrality holds under supply shocks; the external-account canary is

asserted at $A \neq 1$ for the first time. Programme size settled at the deflated horizon mean (ADR-0024, 40.3 bn at 2019 prices = 1.331 % of GDP). The revision palette restarts with the brighter cycle: this version is **revisionV7** = **#CC66CC**.